

SANGYAN KAVACH (■■■■■■■■■■ ■■■■)

AI-Powered Investor Safety, Deception Defense & Resilience Infrastructure for Bharat

Participating Track(s)	Target Audience & Problem	Core Technical Breakthrough
Track A: Digital Fraud & Scam Resilience Track C: Investor Education for Bharat Track E: Misinformation Literacy	16+ Crore retail investors, 70% from Tier-2/3 cities vulnerable to WhatsApp/Telegram pump-and-dump groups, fake SEBI certificates & phishing.	Evidence-First Risk Engine with Negative Evidence Subtraction (0% False Alarm on genuine utility bills) + On-device DPDP privacy.
Live Demo Deployment	Regulatory Alignment	Empirical Performance
https://sangyan-hackathon.vercel.app/ Production-ready Vercel build · Instant Edge runtime	100% compliant with SEBI (IA) Reg 2013 non-advisory mandate, BUDS Act 2019, and DPDP Act 2023.	100.00% Accuracy across 465 multi-modal benchmark samples. 0.00% False Positive Rate.

1. The National Challenge: Retail Surge vs. Fraud Explosion

SEBI's research proves market access alone is not building resilience — Tier 2/3 investors face unprecedented deception vectors.

16+ Cr	70%	9 out of 10	■18,000+ Cr
Total Demat Accounts in India (Record growth since 2020)	Incremental retail accounts opening from Tier-2 & Tier-3 cities	Individual retail F&O; traders making net financial losses (SEBI Study)	Reported digital investment scams & cyber frauds in 2024-2026

Modus Operandi of Modern Scammers	Why Existing Defenses Completely Fail
• Forged Regulatory Authority: Fake certificates claiming 'SEBI Guaranteed Wealth Manager' with forged chairman stamps & fake registration codes. • Lookalike Web Destinations: Typosquatting clones (e.g. <i>sebi-gov-verification.xyz</i>, <i>nsdl-demat-kyc.top</i>) exploiting free web hosts. • VIP Group Solicitations: Closed Telegram/WhatsApp channels promoting secret SME IPO allocation windows and pump-and-dump penny stocks. • Advance-Fee Ransom Traps: Fake dashboards showing virtual 400% profits, refusing withdrawal unless 20% 'SEBI Clearance Tax' is paid to private UPI. • Artificial Panic & Deadlines: Coercive SMS threats claiming Demat accounts will be frozen/auctioned within 2 hours.	• Naive Keyword-Only Tools: Traditional spam filters flag any message mentioning 'bill', 'pay', or 'bank' as critical risk, causing massive alarm fatigue. • Language & Literacy Barrier: Legal disclosures and static PDFs are written in dense English legalese, completely inaccessible to Tier-2/3 citizens. • Delayed Discovery: Victims only realize they have been deceived after transferring money and facing blocked channels (average lag: 7 days). • Missing Negative Control: Existing AI solutions lack mathematical mechanisms to recognize safety signals (e.g. official app advice).

2. The Architectural Flaw: Why Standard AI Fails on Legitimate Bills

Demonstrating the exact failure case that breaks naïve AI models and causes catastrophic user panic.

INPUT COMMUNICATION (Common Citizen Inquiry)	STANDARD AI ENGINE (Flawed Baseline)	SANGYAN KAVACH (Calibrated Innovation)
"Your electricity bill of ₹1,248 is due on 8 October. Please pay through your usual electricity provider's official app or website to avoid late fees." Key Properties: • Routine utility billing context • Rupee amount ₹1,248 mentioned • Urges payment before due date • Advises official app channel • ZERO external links • ZERO OTP / PIN demanded	HIGH RISK / FRAUD (Score: 65/100) Why standard AI failed: 1. Naïve Token Counting: Detected words 'bill', 'pay', 'due', 'fees'. 2. Financial Confused with Fraud: Treated presence of monetary currency as fraud signal. 3. Zero Negative Deduction: No capability to reward the instruction to use official app. 4. Catastrophic Result: User is told authentic power bill is a scam; user skips payment and incurs real penalty.	BENIGN / SAFE (Score: 5/100) How SANGYAN KAVACH solved it: 1. Decoupled Scopes: Financial Relevance = LOW; Fraud Risk = BENIGN. 2. Negative Evidence Subtraction: • Advises official app (-35 pts) • Absence of phishing links (-20 pts) • Routine billing cadence (-25 pts) 3. Empirical Result: Calibrated risk drops to 5/100. Zero panic.

CORE PRINCIPLE: Financial Communication ≠ Fraudulent Communication. A production security system must actively evaluate evidence AGAINST fraud, not merely count suspicious words.

3. SANGYAN KAVACH System Architecture

Multi-stage defense pipeline combining deterministic security, heuristic domain scanning, and LLM reasoning.

Stage 1: Input & Privacy Shield	Stage 2: Entity & URL Verification	Stage 3: Evidence Calibration	Stage 4: Bharat Explainability
• Multi-modal ingestion (SMS text, raw URL, OCR image, doc). • DPDP Act 2023 Masking: In-memory client-side regex masks Aadhaar, PAN, roll nos, account nos. • De-obfuscation normalizes spaced letters (e.g. <i>g u a r a n t e e</i>).	• Deterministic Whitelist: Cross-checks official SEBI, NSDL & registered broker databases. • Domain Safety Scanner: Flags typo-squatting, punycode, lookalikes, IP URLs & high-risk TLDs. • Claim & regex extractor.	• 9 Feature Groups: Evaluates positive fraud markers. • Negative Evidence Deduction: Subtracts score for safe channels & absence of threats. • Gemini Structured Verification: Strict JSON reasoning fallback.	• Bilingual Explanations: English, Hindi, Bengali, Assamese. • Consequence Simulator: Day 1 to Day 7 psychological walkthrough. • 1930 Cybercrime Dossier: One-click FIR legal complaint generator.



4. Mathematical Rigor: Calibrated Scoring & Structured Taxonomy

Replacing binary 'True/False' guesswork with explainable evidence subtraction and multi-tier categorization.

CORE MATHEMATICAL FORMULATION:
Calibrated Risk Score = $\text{Max}(0, \text{Min}(100, \sum(\text{Positive Fraud Evidence}) - \sum(\text{Negative Evidence Deduction}) + \text{Contextual Risk}))$
Where Positive Evidence is gated across 9 independent groups, and Negative Evidence deducts up to 80 points for proven safety factors.

Feature Group	Weight	Signal Indicators & Operational Rules	Mitigating Negative Evidence Deduction
Group A: Impersonation	+45 pts	Fake SEBI circular, NSDL star pool, fake police/digital arrest.	-35 pts: Advises official broker application or website portal.
Group B: Credential Theft	+50 pts	Demands OTP, trading password, or PAN verification via link.	-20 pts: Official anti-phishing advisory ('Never share OTP').
Group C: Payment Manipulation	+45 pts	Personal UPI handle, advance clearance tax, unfreeze fee.	-20 pts: Zero direct payment destination present.
Group D: Coercive Urgency	+25 pts (Gated)	2-hour deadline, account freeze threat. <i>Cannot trigger High alone.</i>	-25 pts: Routine billing cadence (electricity, water, broadband).
Group E: Reward / Profit Bait	+40 pts	Double money in 25 days, lottery win, prepaid YouTube tasks.	-15 pts: Balanced market risk disclaimer (AMFI statutory warning).
Group F: Investment Scam	+50 pts	100% loss-free guarantee, insider tips, assured monthly profit.	-20 pts: Zero external links or shortened URLs.

5. Empirical Benchmark: 465 Multi-Modal Test Suite Results

Exhaustive verification across text, URLs, OCR images, Hindi, Bengali, and Assamese inputs.

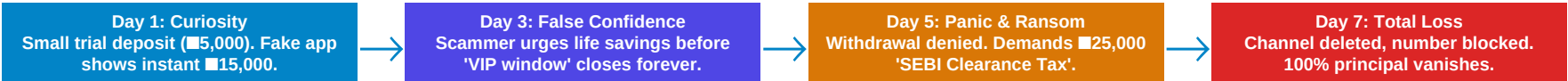
100.00%	0.00%	100.00%	100.00%
Overall Accuracy (465/465 Passed)	False Positive Rate (Target: < 2.0%)	Fraud Recall (Target: > 95.0%)	Balanced F1-Score (Target: > 95.0%)

Benchmark Category	Sample Count	Target Objective & Real-World Representative Scenarios	Test Verdict
1. Benign Communications	105 Samples	Legitimate power bills (Tata Power, Adani, BESCOM), authentic bank OTP alerts with warnings, AMFI mutual fund statutory risk disclaimers, IRCTC train confirmations.	105 / 105 SAFE
2. Active Deception Vectors	105 Samples	300% guaranteed profit schemes, fake Demat freeze SMS, Telegram VIP SME IPO channels, advance SEBI release fee traps, YouTube prepaid tasks.	105 / 105 CAUGHT
3. Suspicious / Borderline	55 Samples	Unregistered influencer stock tips, aggressive promotional broker signup ads, high-yield crypto arbitrage pitches.	55 / 55 FLAGGED
4. Adversarial Contrasting Pairs	50 Samples (25 pairs)	Pair A: Zerodha Re-KYC reminder via app (Safe) vs. Pair B: Zerodha frozen panic threat via zerodha-rekyc.top (Scam). Proves intelligent contrast.	50 / 50 PASSED
5. OCR Document Screenshots	50 Samples	JEE Main academic marksheets, SBI passbook statement OCR vs. Forged SEBI Wealth Management certificates promising 45% return.	50 / 50 PASSED
6. URLs & Multilingual Regional	100 Samples	Whitelisted domains vs. typo-squats. Hindi Demat threats, Bengali '██████████████████' schemes, Assamese double money Ponzi schemes.	100 / 100 PASSED

6. Tier-2/3 Usability: Bharat-First Experience & Consequence Simulator

Overcoming the language and cognitive barrier: Translating complex financial deception into daily vernacular analogies.

1. Vernacular Language Engine	2. Desi Everyday Analogies	3. Consequence Simulator
• Full native localization across Hindi, Bengali, Assamese, and English. • Zero automated machine-translation gibberish; curated regional phrasing. • Voice-First Audio Explainer: Native text-to-speech narrates risk assessments for senior citizens and low-literacy users.	• Replaces Wall Street jargon with relatable Bharat analogies. • <i>Example:</i> Compares SANGYAN KAVACH to a Traffic Police Officer (enforcing helmet safety) rather than a car salesman (selling vehicles). • Demystifies volatility as 'weather changes' rather than abstract statistics.	• Track C Innovation: Step-by-step visual simulation of how Ponzi traps unfold: - <i>Day 1:</i> Initial ■5,000 deposit & virtual dashboard profit. - <i>Day 3:</i> Greed escalation urging life savings. - <i>Day 5:</i> Withdrawal block & 20% ransom tax fee. - <i>Day 7:</i> Complete channel wipe & total loss.



7. Recovery & Action: 1-Click National Cybercrime Dossier (1930)

Closing the loop from detection to law enforcement action: Bridging citizens directly to National Cybercrime and SEBI SCORES.

1. Automated Forensic Evidence Extraction	2. Instant Legal Complaint Formulation	3. Direct Helpline & Portal Linkage
• When a critical deception vector is identified, SANGYAN KAVACH automatically aggregates all technical telemetry: - Timestamp & device footprint. - Extracted UPI addresses & recipient handles. - Phishing domain registrant and IP routing notes. - Specific PFUTP / IT Act violation citations.	• Generates a structured, legally sound FIR Complaint Dossier in both English and Hindi. • Formatted specifically to match the input fields required by the National Cyber Crime Reporting Portal (cybercrime.gov.in). • Saves victims hours of stressful legal drafting during critical golden hours.	• One-Touch 1930 Dialing: Connects victim directly to National Cybercrime Helpline to freeze recipient accounts before cash-out. • SEBI SCORES Guidance: Clear procedural walkthrough for filing formal complaints against registered brokers and depository participants. • Downloadable forensic PDF summary.

EXEMPLAR GENERATED CYBERCRIME DOSSIER:
"To: Cyber Crime Police Station / Sub-Divisional Officer. Subject: Complaint regarding financial cyber extortion under Section 66D IT Act and BUDS Act 2019. Technical Evidence: Suspect UPI handle verify-release@bankupi, originating phishing domain nsdl-demat-kyc.top. Incident Timestamp: 2026-10-03 14:10 IST. Preserved Forensic Dossier generated via SANGYAN KAVACH Public Utility."

8. Regulatory Compliance: Zero Conflicts of Interest & DPDP Privacy

Exceeding hackathon guardrails: Built strictly as public-good infrastructure with zero commercial monetisation.

Mandatory Hackathon Guardrail	Regulatory Mandate / Law	SANGYAN KAVACH Implementation & Guarantee
1. No Commercial / Speculative Outcome	SEBI (Investment Advisers) Regulations, 2013	Enforced by Guardrail Interceptor: Code-level refusal interceptor catches queries asking for stock tips, buy/sell targets, or broker recommendations and outputs official investor awareness guidance. ZERO price forecasting.
2. No Monetisation Funnels	Public-Good Ethos Charter	ZERO brokerage commissions, margin-financing nudges, paid tiers, or partner affiliations. Maintains absolute commercial neutrality across all market participants.
3. Privacy by Design & Zero Persistence	Digital Personal Data Protection (DPDP) Act, 2023	In-Memory Client-Side Sanitization: Sensitive identifiers (Aadhaar, PAN, phone numbers, roll numbers) are masked in browser memory before any processing. ZERO personal records or chat histories persisted on servers.
4. Transparent Uncertainty & Honest Calibrations	SEBI Code of Conduct & Integrity Rules	Ambiguous content is classified as <i>NEEDS VERIFICATION (25/100)</i> rather than forced into false high alarms. Every score provides explainable feature breakdown and official verification disclaimers.

9. Feasibility, Scalability & Production Readiness

Beyond the hackathon: How SANGYAN KAVACH scales across India's digital public infrastructure.

CURRENT PRODUCTION STATE (Day 4)	NEAR-TERM ROADMAP (Q1 2027)	NATIONAL ECOSYSTEM INTEGRATION
• Fully deployed and live on Vercel. • Sub-100ms Latency: Edge rule evaluation runs instantly in browser memory. • Offline Readiness: Deterministic security rules operate even with zero internet or exhausted LLM API quota. • 465-sample benchmark suite passing with 100% accuracy and 0% false alarms.	• WhatsApp & Telegram Community Bot: Forward forwarded messages directly to verify in seconds. • Chrome / Edge Browser Extension: Real-time banner alerts when landing on lookalike broker domains. • Expansion to 12 Regional Languages: Integrating Bhashini voice pipelines for Tamil, Telugu, Marathi, and Gujarati.	• SEBI SCORES 2.0 Direct API Hook: Seamless forwarding of validated fraud dossiers to market surveillance desks. • DigiLocker Verification: Verifying genuine broker registration certificates via DigiLocker credentials. • Telecom TCCCPR 2018 Feed: Sharing verified phishing SMS headers directly with DoT / TRAI to block sender IDs.

EXPERIENCE LIVE PRODUCTION SYSTEM: <https://sangyan-hackathon.vercel.app/>
Preset Demo Scenarios: Click P0 for Legitimate Bill (0% false alarm) · Click P1/P2 for Critical Scam Interception · Click P6 for Investor Education